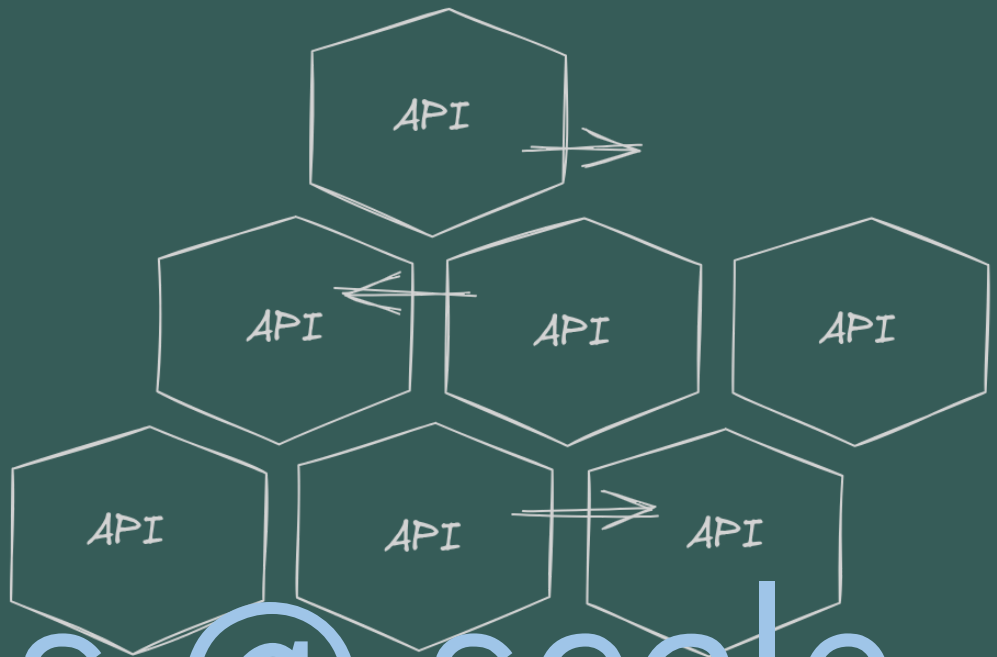


Hateoas @ scale

Hateoas implementation in microservices architecture

What is Hateoas ?

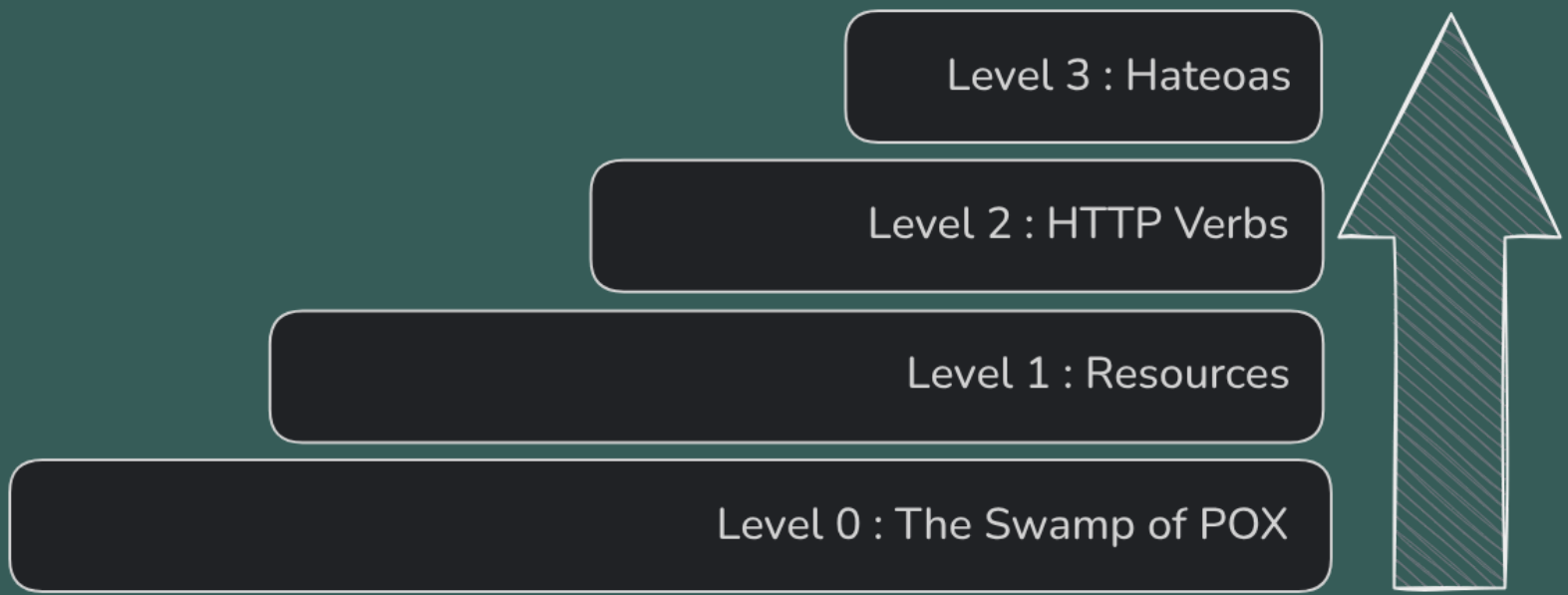
What is Hateoas ?

Hypermedia As The Engine Of the Application State

What is Hateoas ?

Hypermedia As The Engine Of the Application State

Richardson model of REST maturity

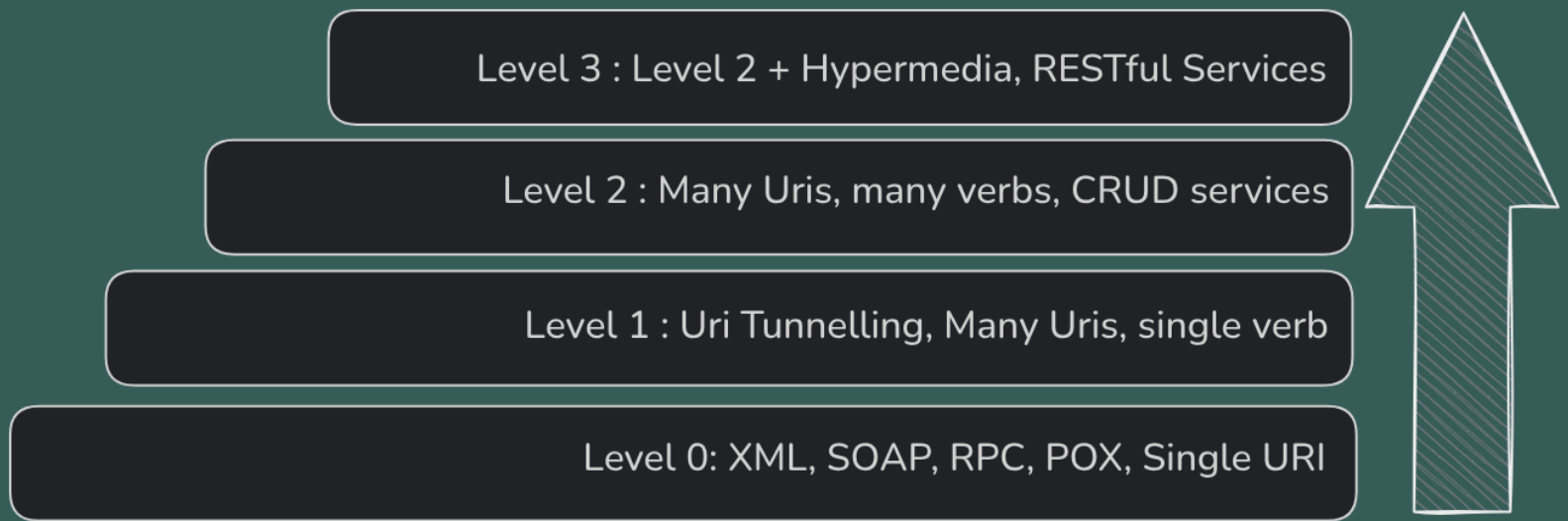


source: Martin Fawler

What is Hateoas ?

Hypermedia As The Engine Of the Application State

Richardson model of REST maturity

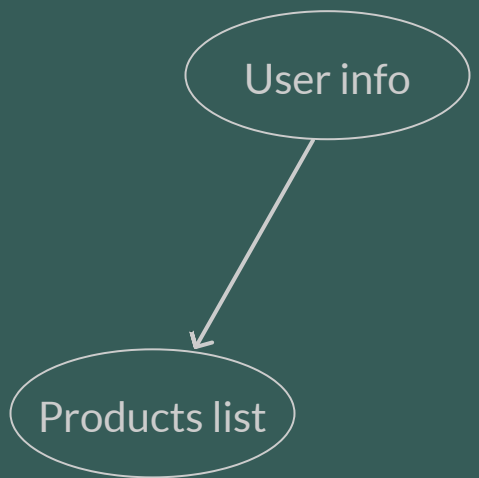


source: Martin Fawler

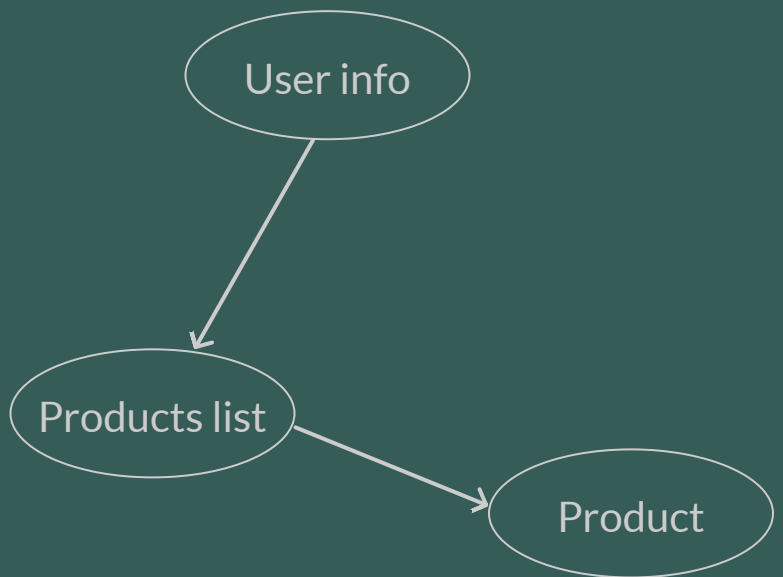
Marketplace User Workflow



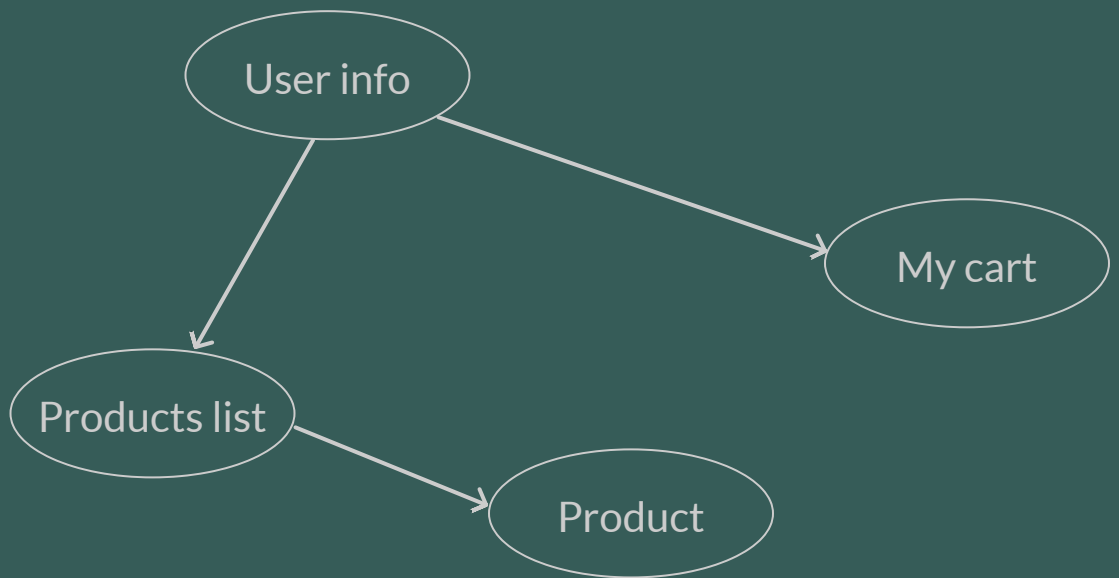
Marketplace User Workflow



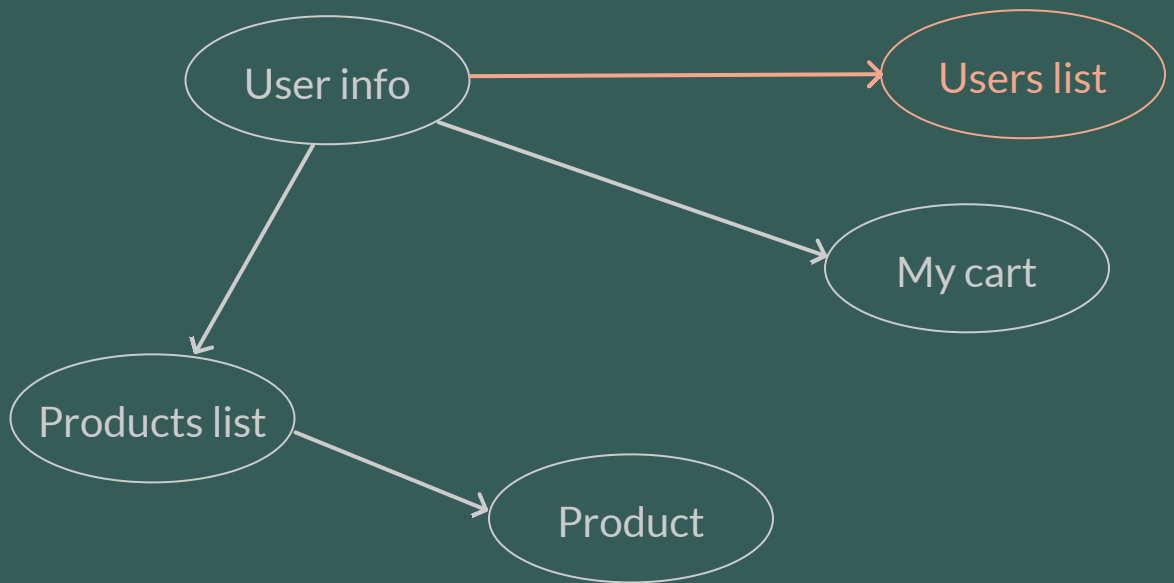
Marketplace User Workflow



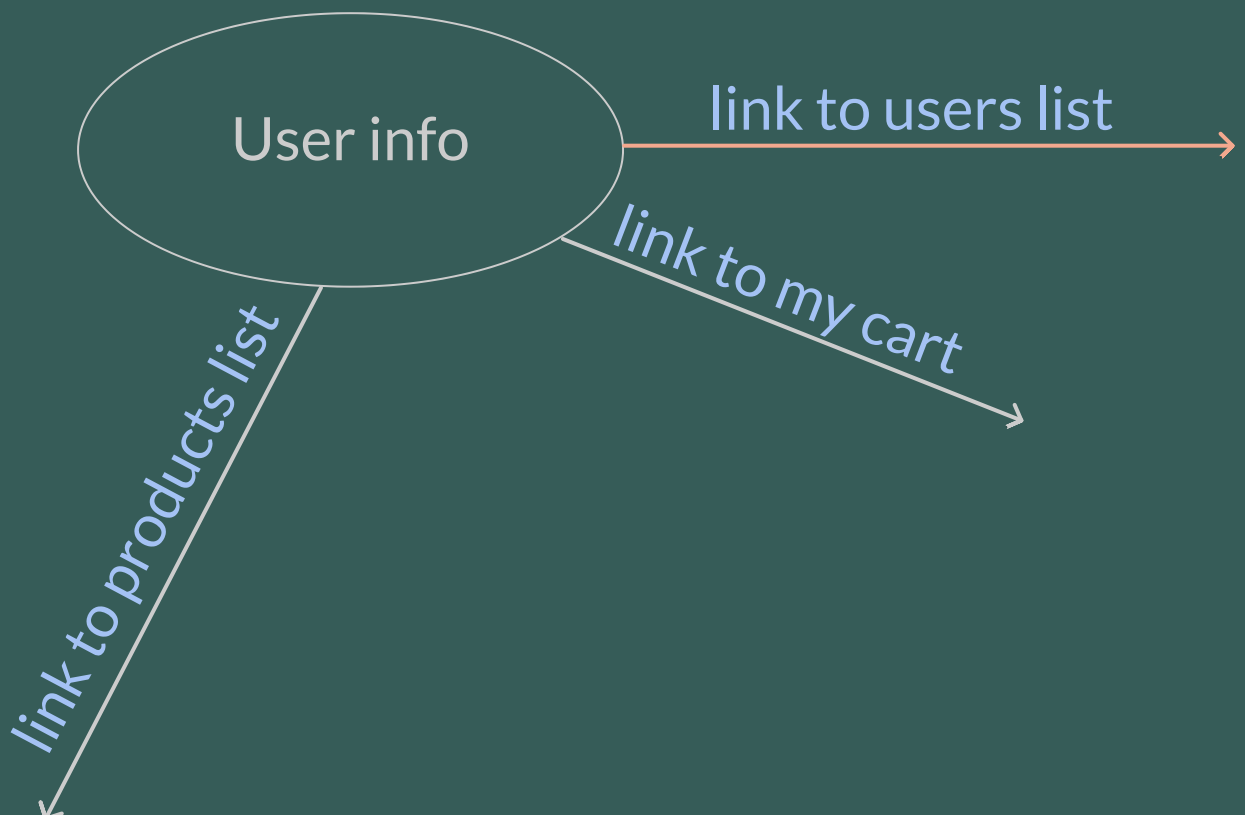
Marketplace User Workflow



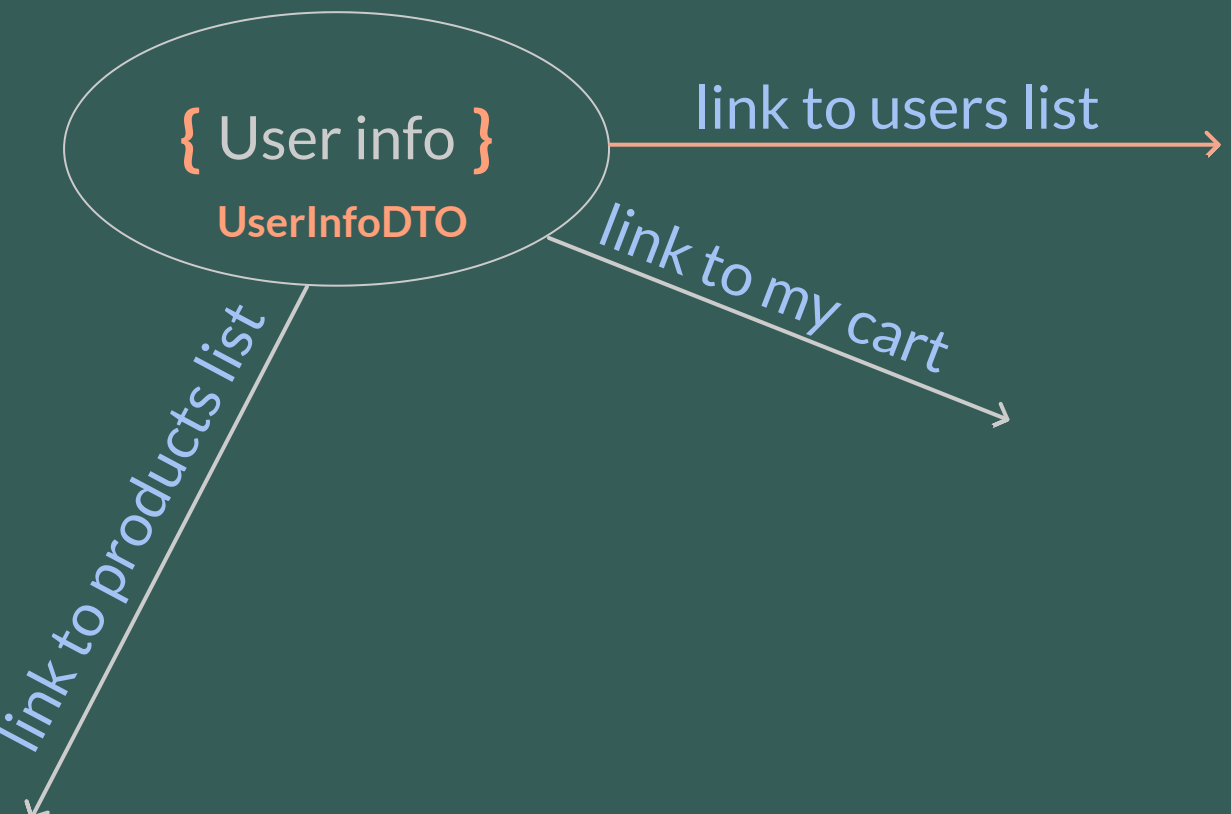
Marketplace User Workflow



Marketplace User Workflow

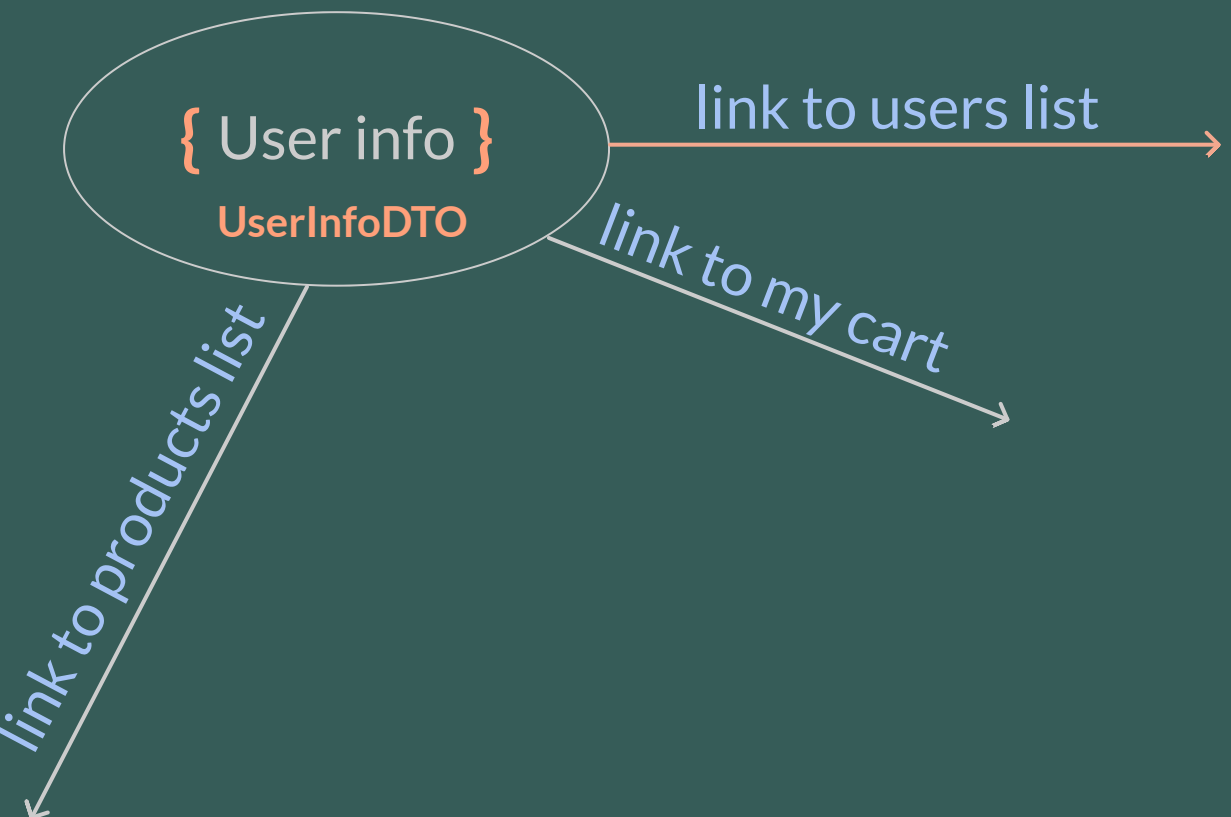


Marketplace User Workflow



Marketplace User Workflow

/api/v1/users/userinfo



UserInfoDTO

/api/v1/users/userinfo

```
{
    "username": "ada.lovelace",
    "firstname": "Ada",
    "lastname": "Lovelace"
}
```

Workflow DTO

/api/v1/users/userinfo

```
{
    "username": "ada.lovelace",
    "firstname": "Ada",
    "lastname": "Lovelace"
}
```

Workflow DTO

/api/v1/users/userinfo

```
{
    "username": "ada.lovelace",
    "firstname": "Ada",
    "lastname": "Lovelace"
}
```

/api/v1/products

```
{
    "list": [
        {
            "name": "pineapple",
            "price": 20.04
        },
        ...
    ]
}
```

Workflow DTO

/api/v1/users/userinfo

```
{
    "username": "ada.lovelace",
    "firstname": "Ada",
    "lastname": "Lovelace"
}
```

/api/v1/products

```
{
    "list": [
        {
            "name": "pineapple",
            "price": 20.04
        },
        ...
    ]
}
```

/api/v1/users

```
{
    "list": [
        {
            "username": "ada.lovelace",
            "firstname": "Ada",
            "lastname": "Lovelace"
        },
        ...
    ]
}
```

Workflow DTO

/api/v1/users/userinfo

```
{
    "username": "ada.lovelace",
    "firstname": "Ada",
    "lastname": "Lovelace"
}
```

/api/v1/products

```
{
    "list": [
        {
            "name": "pineapple",
            "price": 20.04
        },
        ...
    ]
}
```

/api/v1/users

```
{
    "list": [
        {
            "username": "ada.lovelace",
            "firstname": "Ada",
            "lastname": "Lovelace"
        },
        ...
    ]
}
```

/api/v1/carts/mycart

```
{
    "totalPrice": 1147.00,
    "user": {
        "username": "ada.lovelace"
    },
    "products": [
        {
            "name": "apple",
            "totalPrice": 10.00,
            "quantity": 1
        }
        ...
    ]
}
```

Workflow DTO with Hateoas

/api/v1/users/userinfo

```
{
    "username": "ada.lovelace",
    "firstname": "Ada",
    "lastname": "Lovelace",
    "_links": {
        "allUsers": {
            "href": "http://localhost:8080/api/v1/users/all"
        },
        "allProducts": {
            "href": "http://localhost:8080/api/v1/products"
        },
        "myCart": {
            "href": "http://localhost:8080/api/v1/cart/my-cart"
        }
    }
}
```

UserInfoDTO

/api/v1/users/userinfo

```
class UserInfoDto @JsonCreator constructor(
    val username: String,
    val firstname: String,
    val lastname: String
): RepresentationModel<UserInfoDto>() {
    init {
        addUsersListLink()
        addProductsListLink()
        addMyCartLink()
    }
...
}
```

UserInfoDTO

/api/v1/users/userinfo

```
1      private fun addUsersListLink() {
2          add(
3              linkTo(
4                  UsersResource::class.java.methods.first { it.name == "findAll" }
5              )
6              .withRel("allUsers"))
7      }
8
9      private fun addProductsLink() {
10         add(
11             linkTo(
12                 ProductsResource::class.java.methods.first { it.name == "findAll" }
13             )
14             .withRel("allProducts"))
15     }
16
17     private fun addMyCartLink() {
18         add(
19             linkTo(
20                 CartsResource::class.java.methods.first { it.name == "myCart" }
21             )
22             .withRel("myCart"))
23     }
```

UserInfoDTO

/api/v1/users/userinfo

```
constructor(
    private val permissions: List<Permission>,
    val username: String,
    val firstname: String,
    val lastname: String
) : RepresentationModel<UserDto>() {
    ...

    private fun addUsersListLink() {
        if (permissions.contains(READ_ALL_USERS)) {
            add(
                linkTo(
                    UsersResource::class.java.methods.first { it.name == "findAll" }
                )
                .withRel("allUsers"))
        }
    }

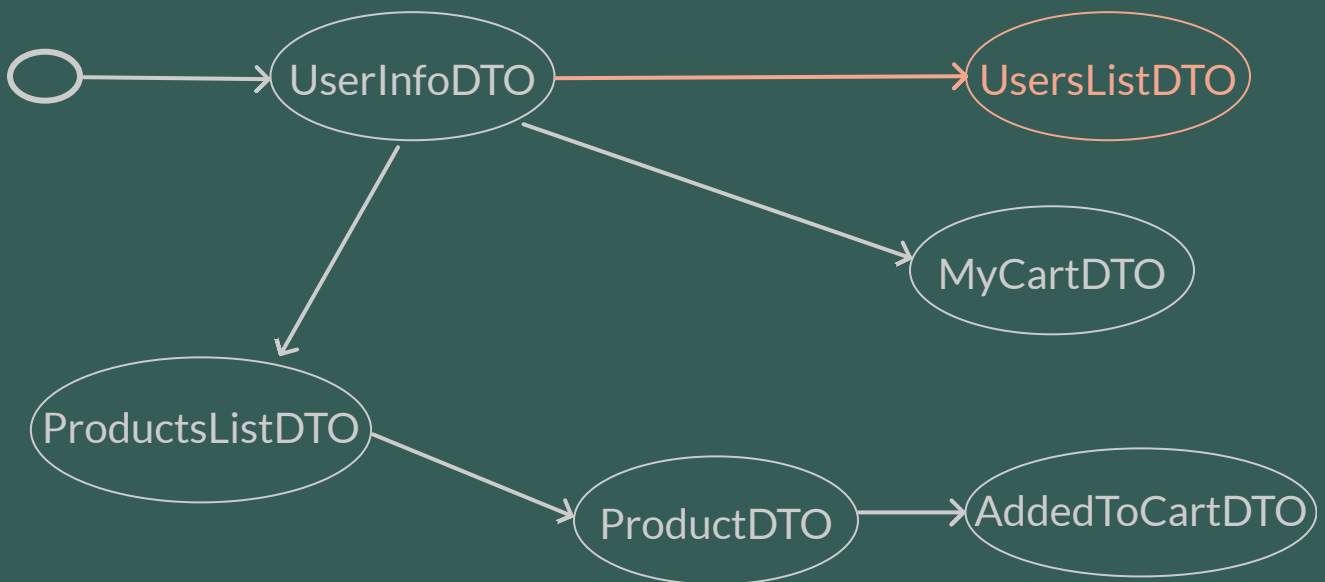
    ...
}
```

State diagram

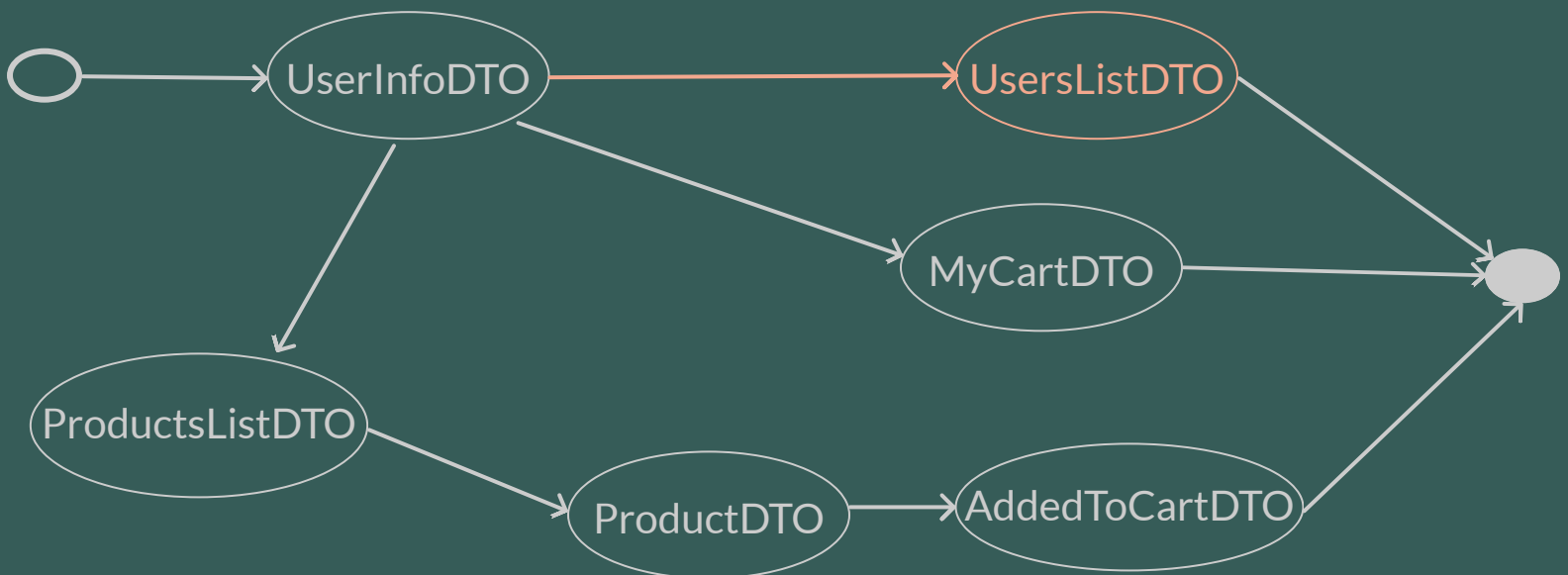
State diagram



State diagram

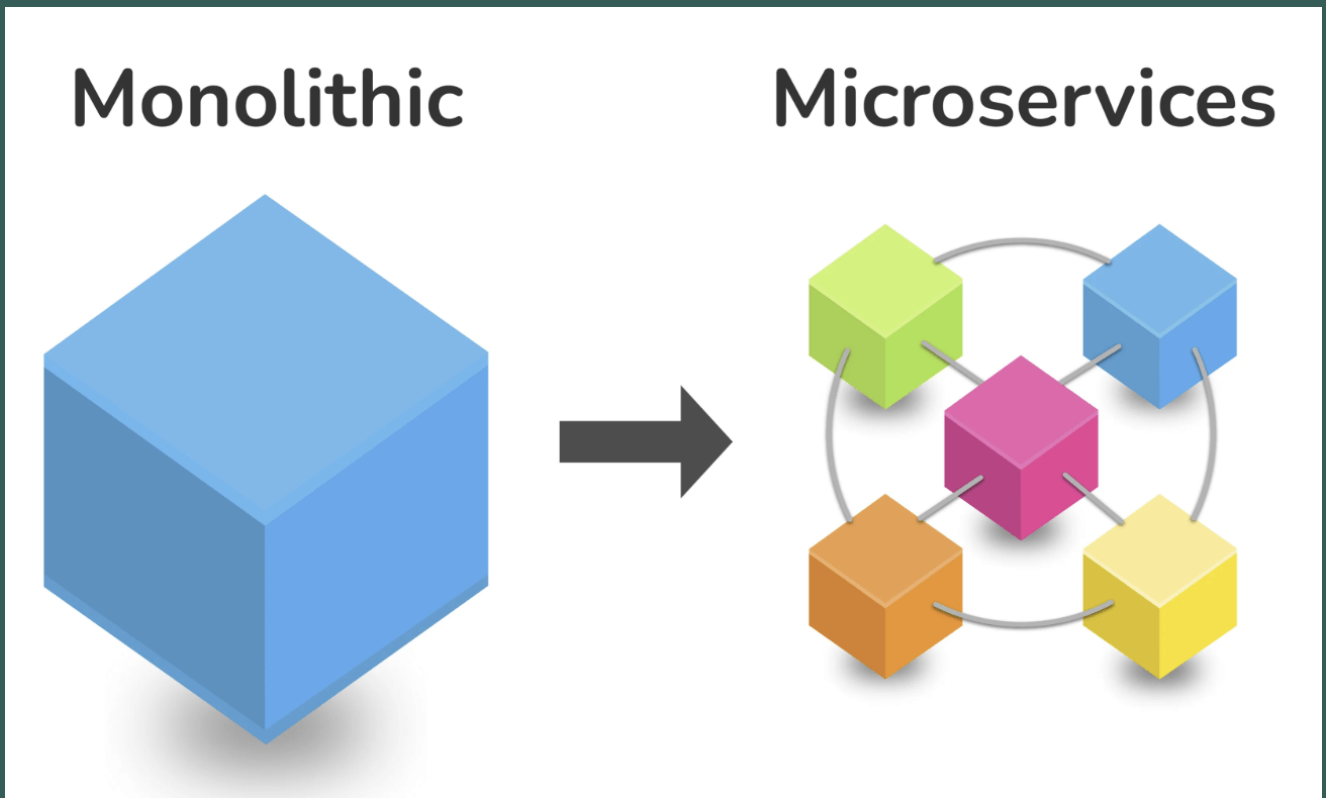


State diagram

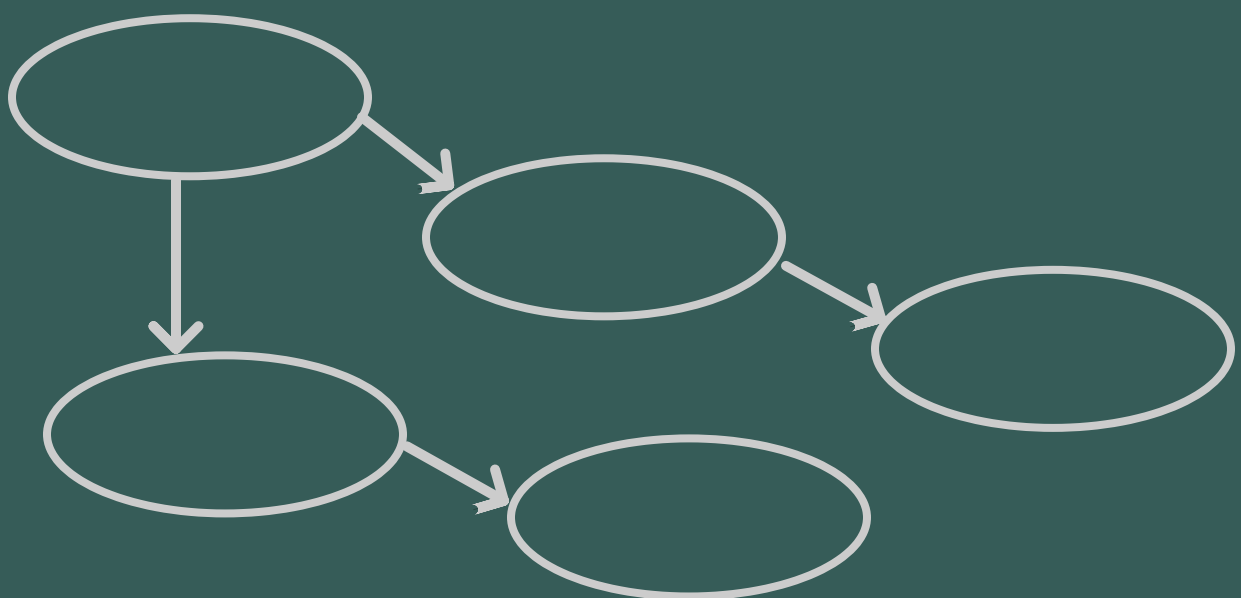


What about microservices ?

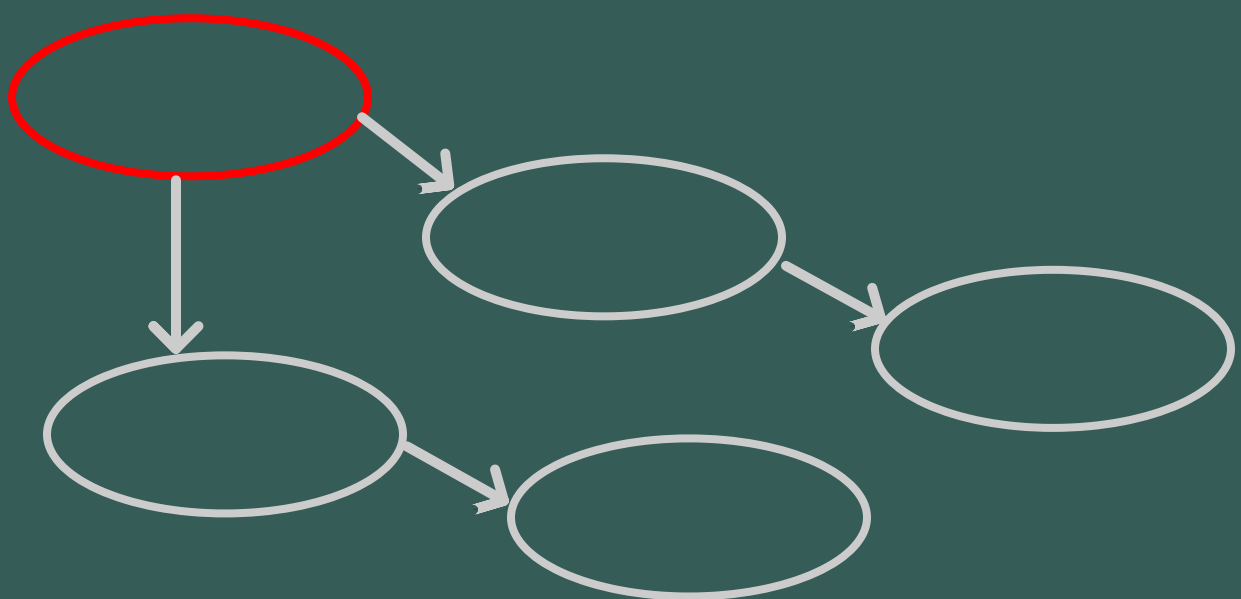
What about microservices ?



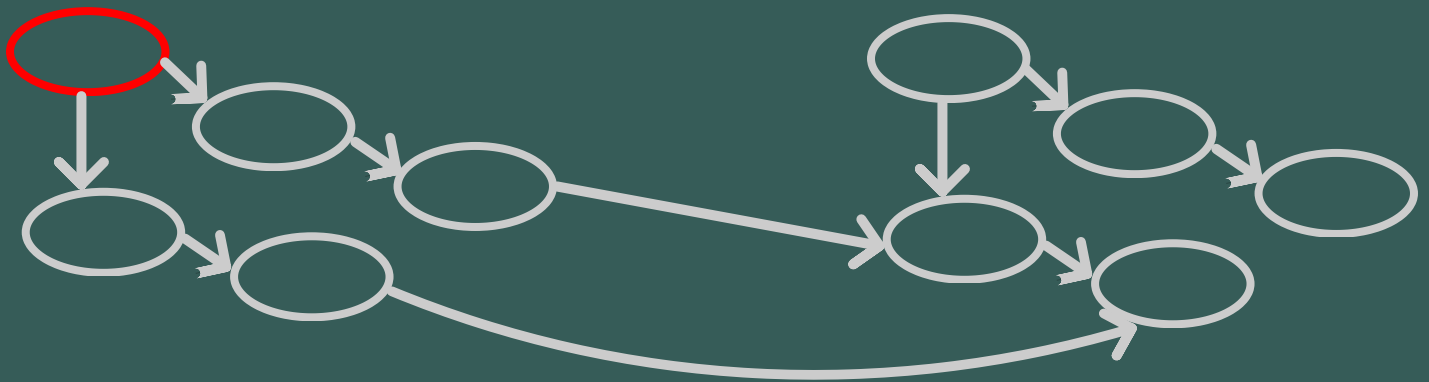
Single workflow



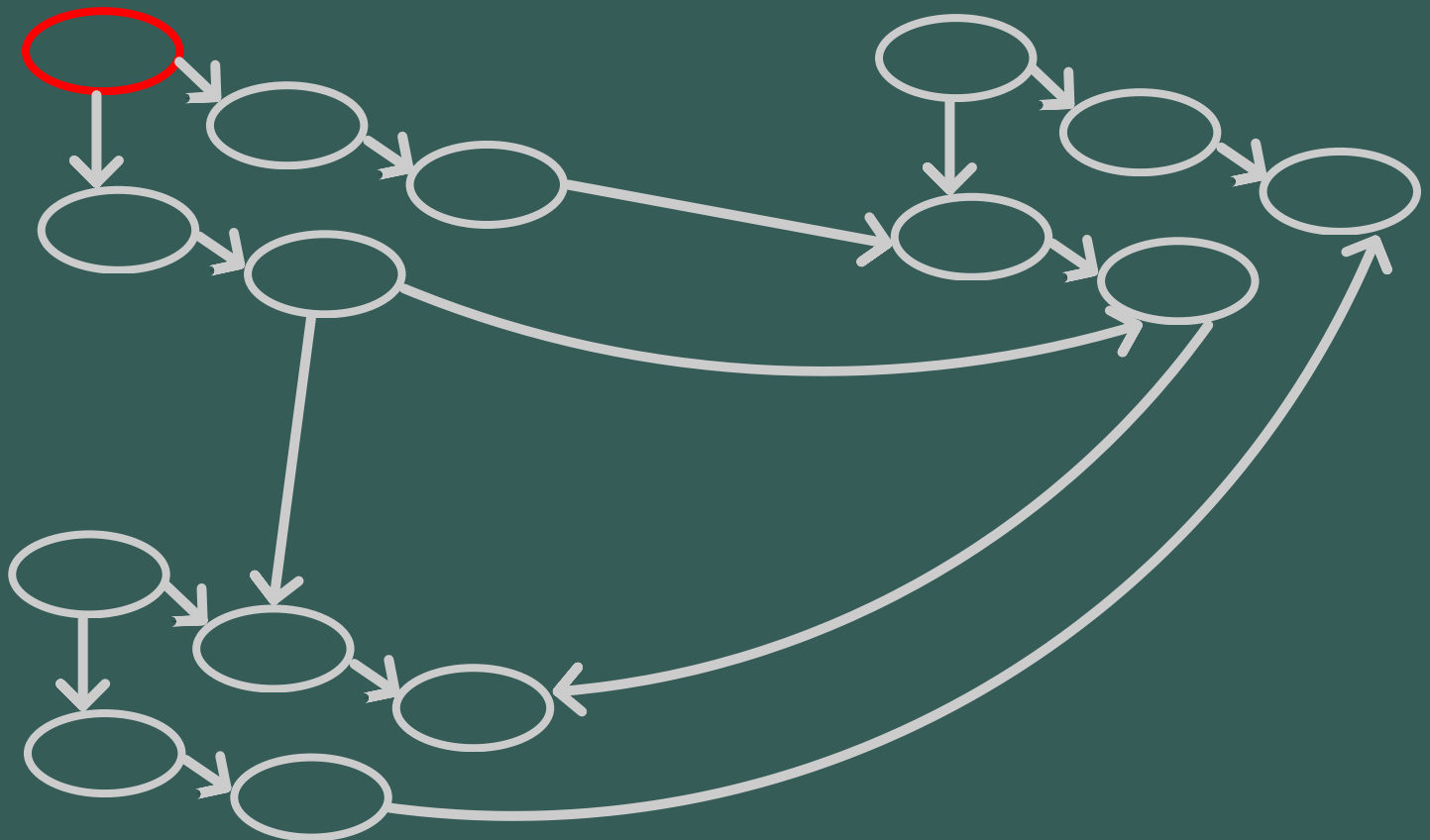
Single workflow



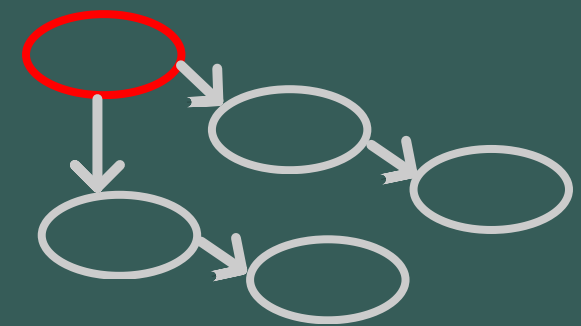
Legacy microservices workflows

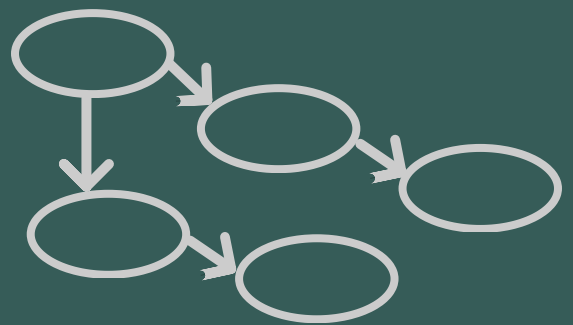


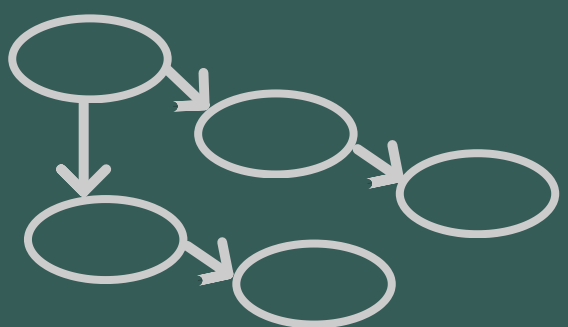
Legacy microservices workflows

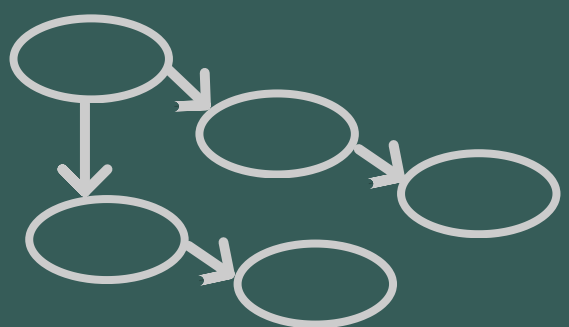


Microservices workflows

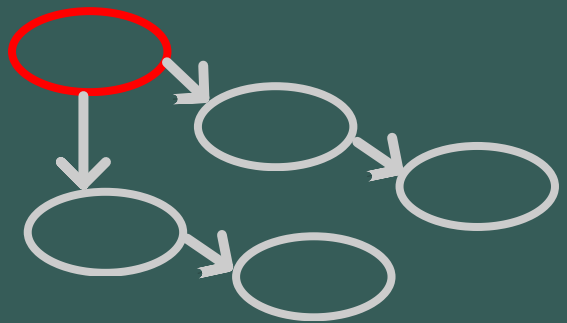


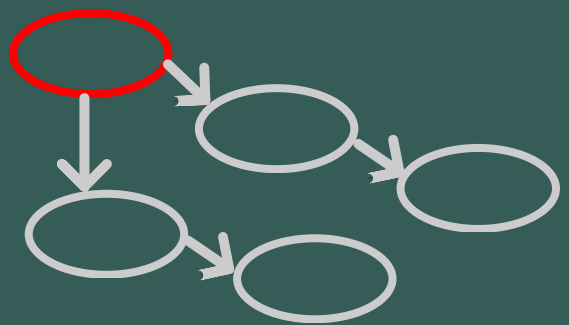


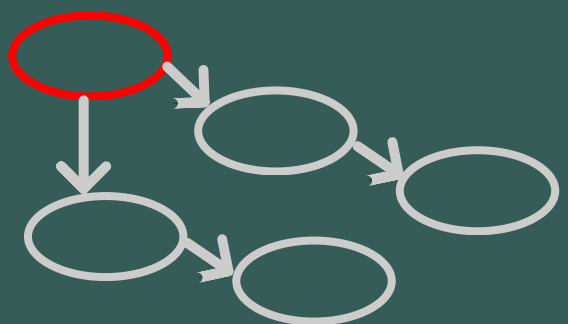


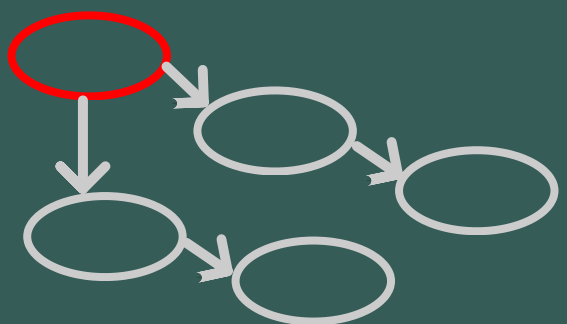


Microservices workflows

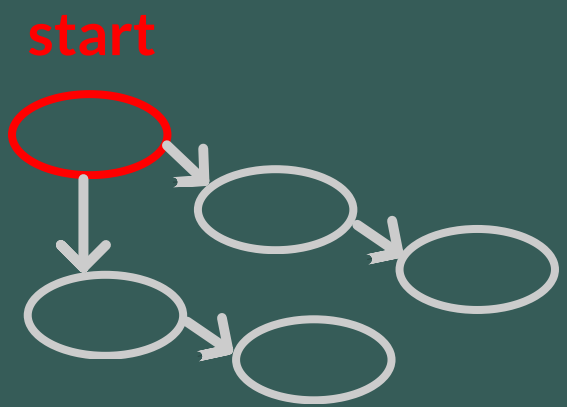


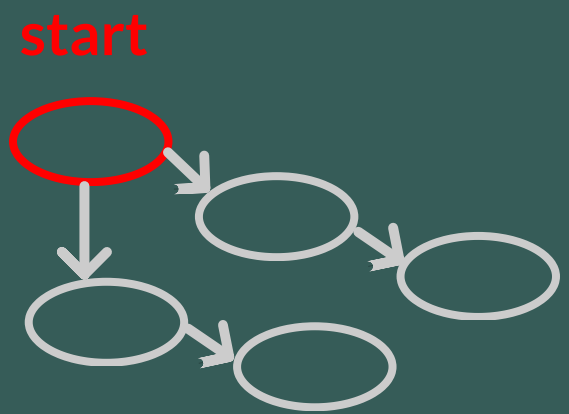


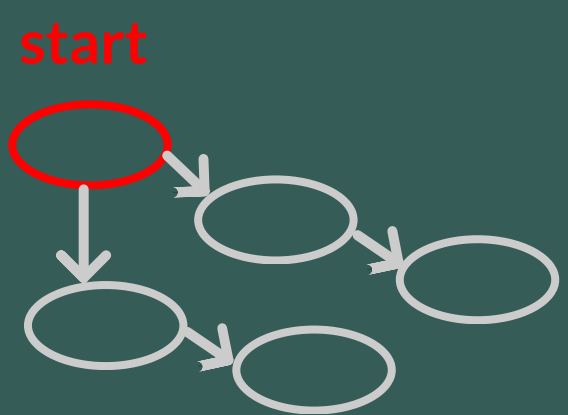


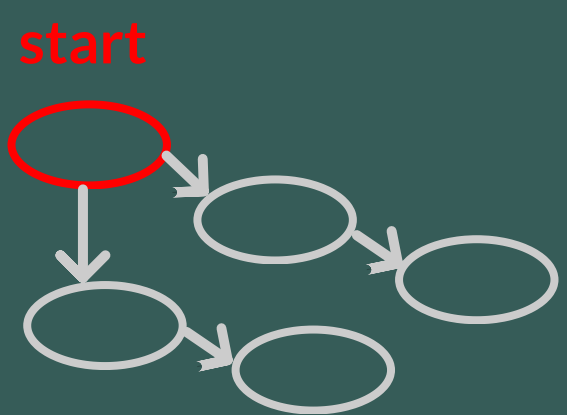


Microservices workflows

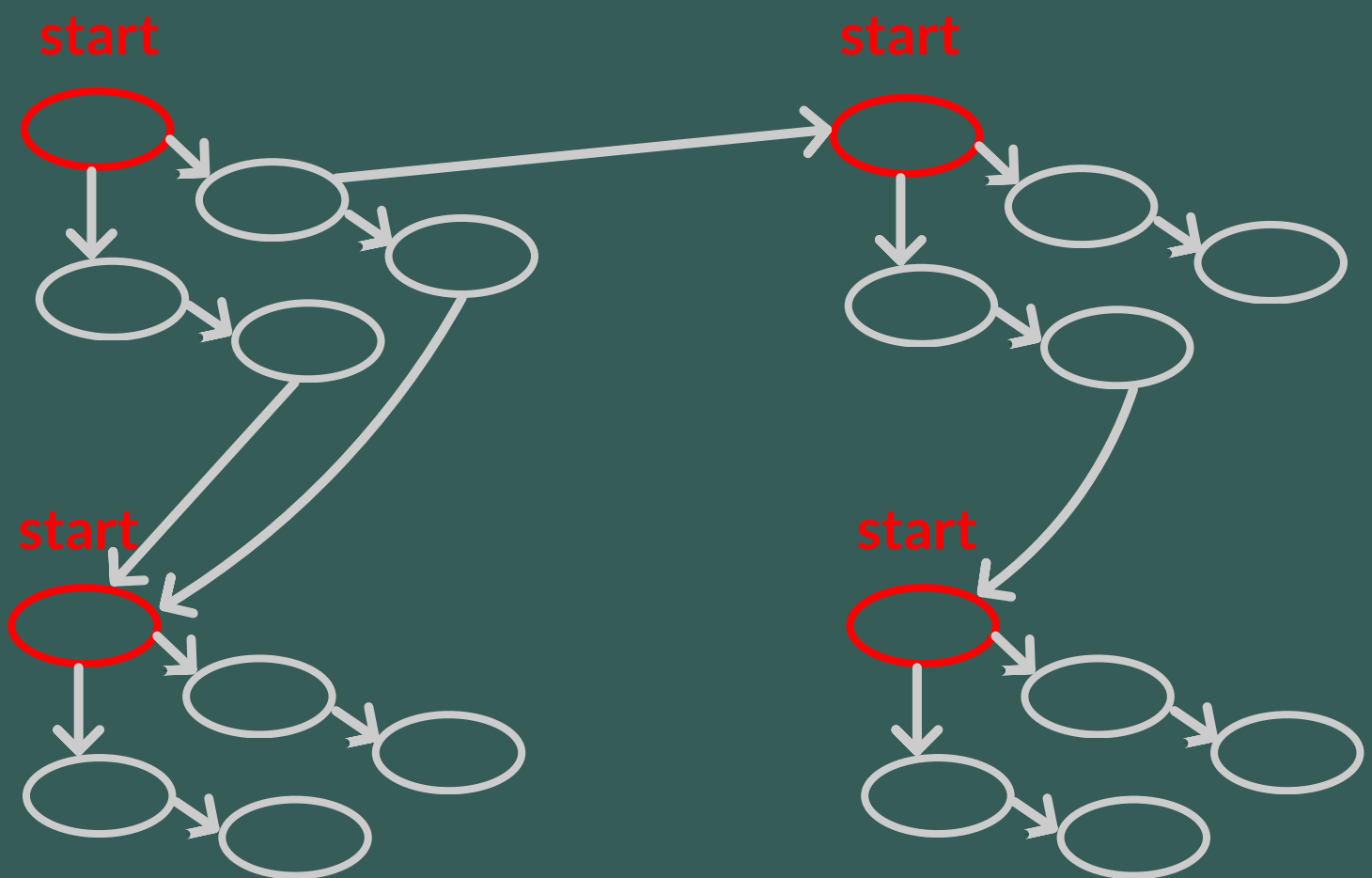




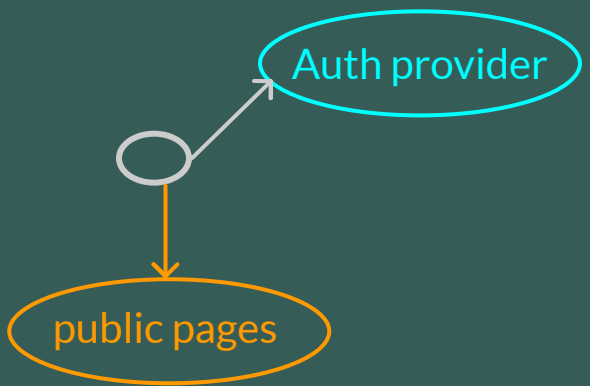




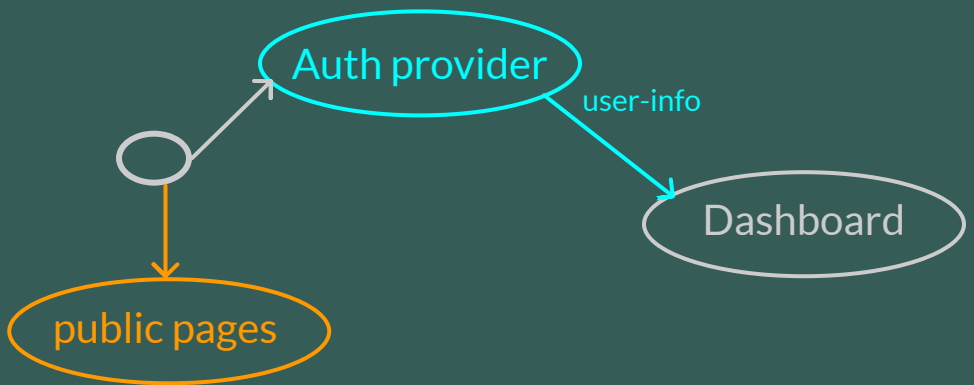
Microservices workflow



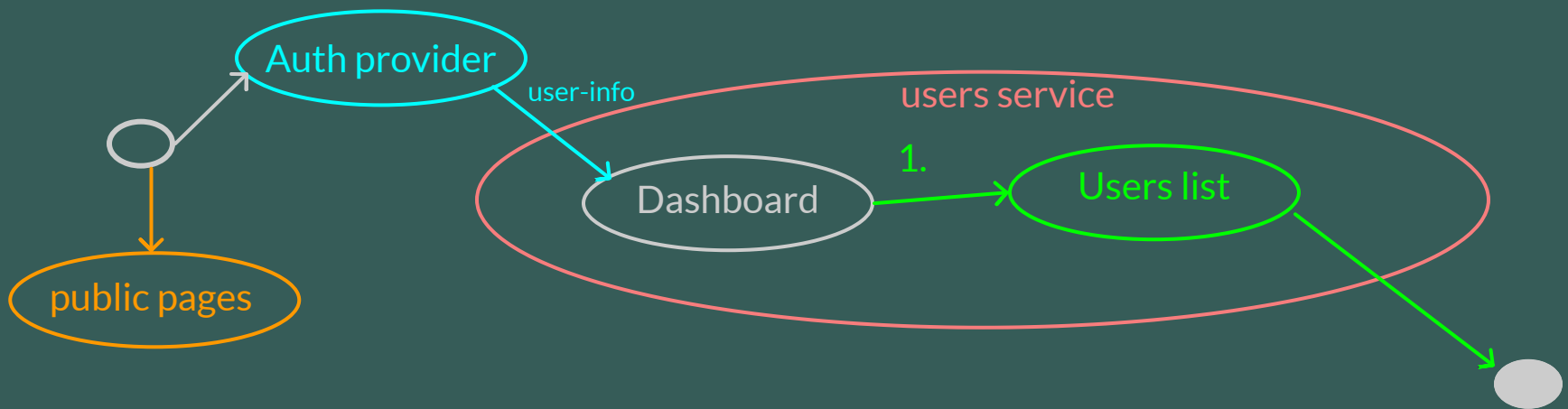
Marketplace workflow



Marketplace workflow



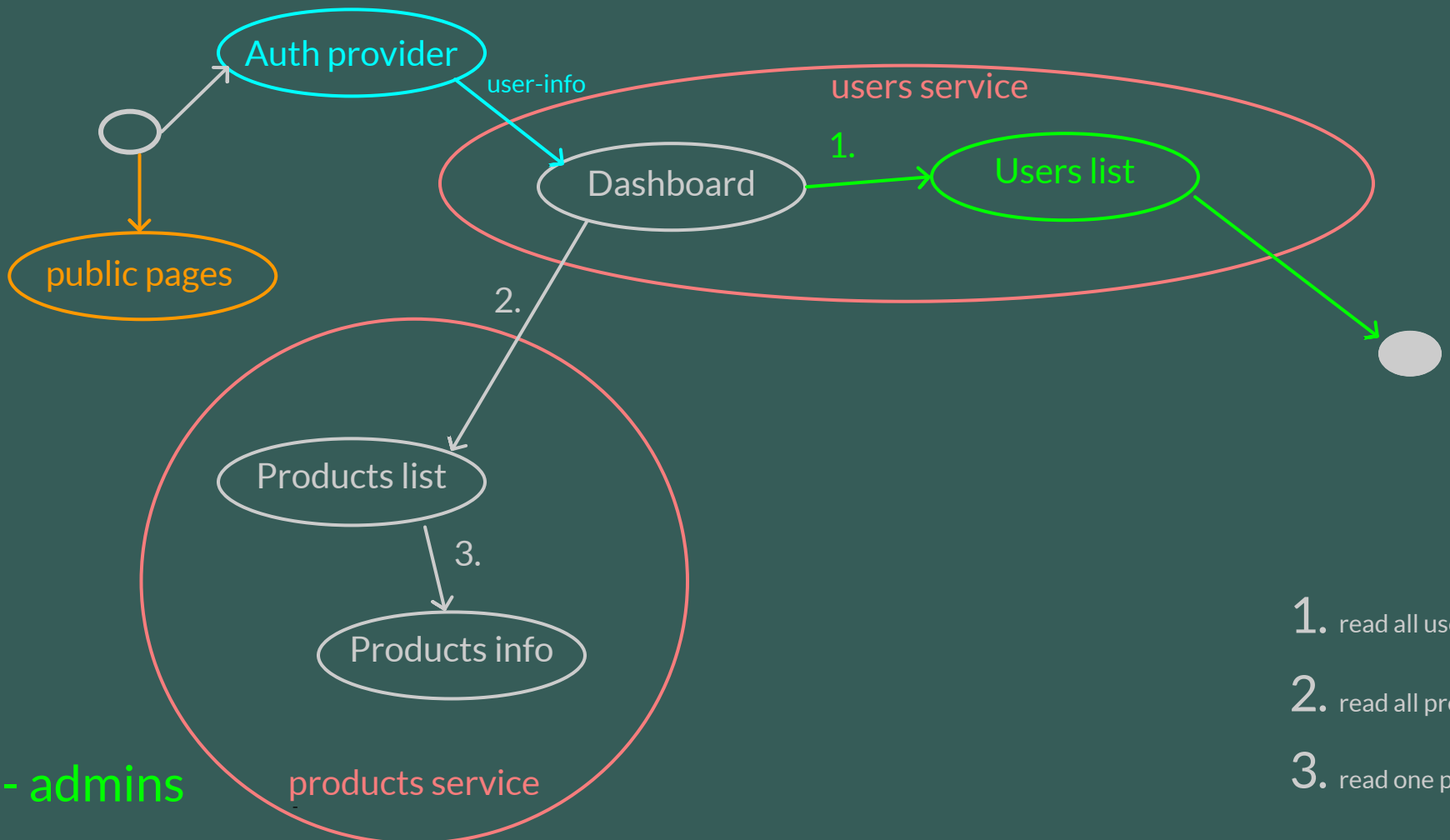
Marketplace workflow



1. read all users

- admins
- customers
- unknown

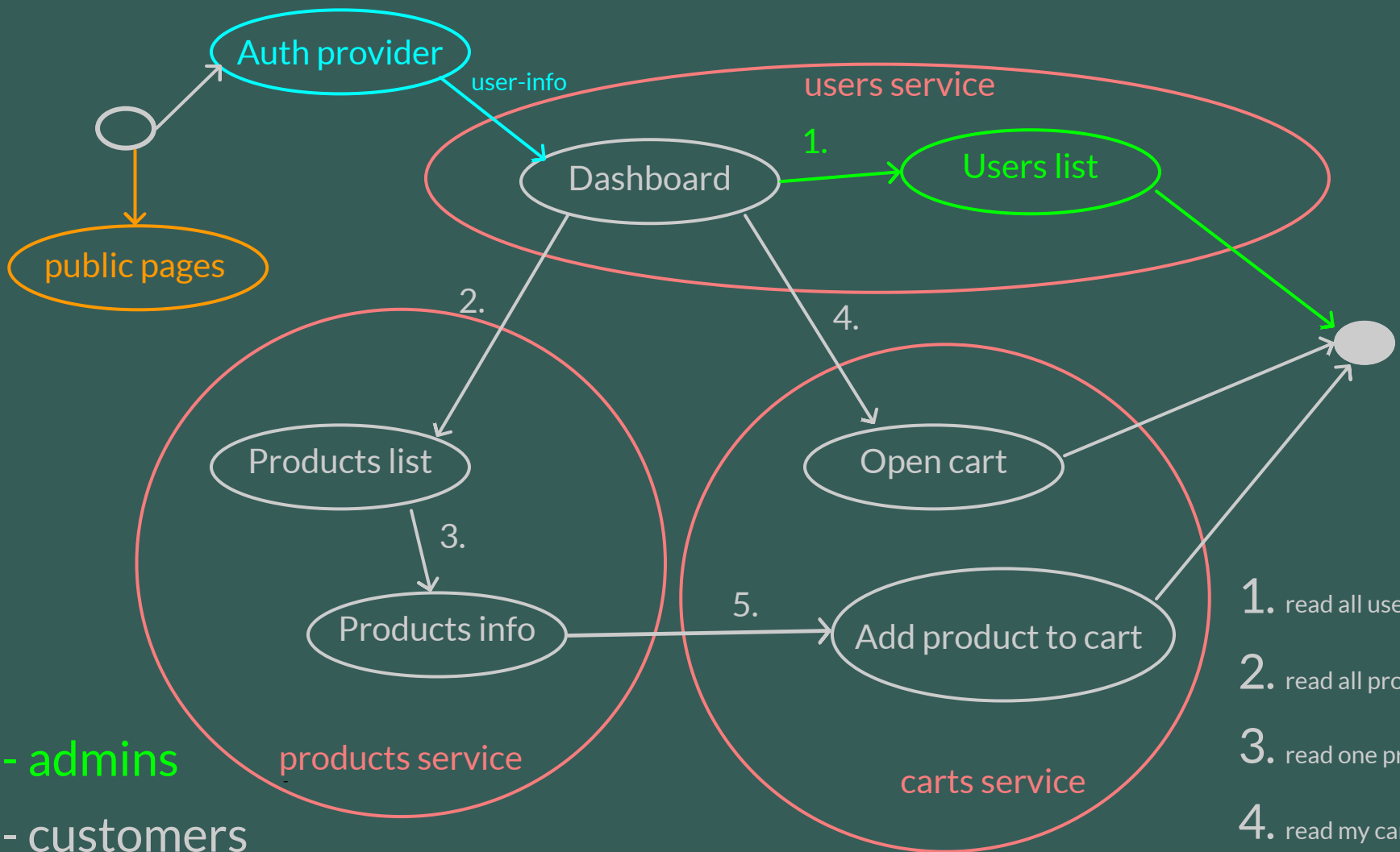
Marketplace workflow



- admins
- customers
- unknown

1. read all users
2. read all products
3. read one product

Marketplace workflow



Implementation

Implementation

